Lecture 15: What Makes an Evaluation Valid?

CS120: Introduction to AI Safety - November 12, 2026, Zachary Robertson

From stakeholder feedback to institutional assurance

Module 2	Module 3	Module 4	Module 5
Model behavior	**Agent policy**	**Interaction process**	**Institutional ecosystem**
shortcut learning	reward hacking	sycophancy	benchmark gaming
miscalibration	specification gaming	feedback-loop amplification	endogenous evaluation

Course sequence, not a hierarchy

Institutions use evaluation evidence to support decisions about AI systems.

Data work sits inside institutions

Commission data work → Interpret evidence → Act on assurance claims

Who organizes the work also shapes what its results are used to justify.

Assurance requires evidence and safeguards

Epistemic assurance

Produce valid evaluations of capabilities and hazards.

Safeguards

Establish and enforce constraints on development and deployment.

An evaluation alone does not establish acceptable residual risk.



Roadmap

What does measurement modeling establish?

When is inference necessary?

What do the diagnostic examples establish?

Observable evidence needs an interpretation

Preference

Choices or rankings

Assumption: Utility axioms apply to revealed choices.

Toxicity

Annotator labels

Assumption: Labels represent the intended notion of toxicity.

Sycophancy

Response given A/B framing

Assumption: The contrast identifies excessive compliance.

Values also enter operationalization

Theoretical understanding

For toxicity: which harms count?

Operationalization

Whose labels count as evidence of those harms?

Values enter both the construct and its operationalization.

Jacobs & Wallach, 2021

A proposed argument for measurement modeling

P1: Many sociotechnical harms result from construct-operationalization mismatches.

P2: Measurement modeling makes these assumptions explicit and some implications testable.

C1: Measurement modeling can help explain, anticipate, and mitigate some sociotechnical harms.

$$P_1 \wedge P_2 \rightarrow C_1$$

Jacobs & Wallach, 2021

Usefulness does not establish necessity

Measurement modeling helps scrutinize a claimed measurement.

The paper asks

What assumptions support the measurement?

Our next question

Does the decision require this construct estimate?

Discussion 1: What exactly is the disagreement?

Choose a reading response and poster name for all three discussions.

Give the strongest reply to the critique in the selected response. Is the disagreement about what the construct means, whether the proxy measures it, or how the score is used?

Which part of the disagreement could additional evidence resolve?

Discuss with classmates; record your own response to Question 1.

Jacobs & Wallach, 2021, secs. 1-2



Roadmap

What does measurement modeling establish?

When is inference necessary?

What do the diagnostic examples establish?

A task can concern a construct without estimating it

Start with the institutional decision: what information does it require?

Would observable information suffice, or must we estimate the construct?

An operationalization does not establish inferability

Observable indicators → Claim about a construct

What warrants the inference?

Specifying indicators does not establish that they support the construct claim.

Reliability is not validity of interpretation and use

Reliability

How consistent is the measurement?

Validity

Does the evidence support this interpretation and use?

A consistent score can still support an invalid interpretation.

Jacobs & Wallach, 2021, sec. 3

First ask whether the task needs inference

Observable

Is there an argument that the construct is unobservable?

Descriptive

Is inferring it necessary to describe or justify the task?

Then ask what the inference explains

Warranted

Do the observable indicators warrant the inference?

Proximate

Is the relevant harm explained by a construct-operationalization mismatch?

Discussion 2: Is this inference necessary?

Keep the same example and focus on the original institutional decision.

Could the institution address the identified harm without estimating the construct? Propose a concrete change based on observable information, or explain why the decision requires that inference.

Would your approach support the original decision, or only address part of the problem?

Discuss with classmates; record your own response to Question 2.



Roadmap

What does measurement modeling establish?

When is inference necessary?

What do the diagnostic examples establish?

Height is observable

Height is "not an unobservable theoretical construct."

Jacobs and Wallach model it as latent for exposition.

Even observable height needs a definition that fits its use.

Jacobs & Wallach, 2021, sec. 2.1

Socioeconomic status (SES) inference

Universal Credit counted two monthly paychecks in one assessment window.

Actual pay: GBP 1,000/month → Assessed income: GBP 2,000/month

Test timing sensitivity by shifting the window. No socioeconomic status (SES) estimate is needed.

Jacobs & Wallach, 2021, secs. 2.2, 3.1.1

A teacher's score changes from 6 to 96

Year 1 → Year 2

6 → 96

Evidence of unreliability if underlying effectiveness remained comparably stable.

Jacobs & Wallach, 2021, secs. 2.3, 3.1

Score instability does not identify its cause

Change

Teacher effectiveness changed.

Noise

Evaluations were noisy.

Bias

Scores were biased.

The score change alone does not distinguish these explanations.

Jacobs & Wallach, 2021, secs. 2.3, 3.1

Explaining harm need not require latent inference

Height: the construct is observable.

SES: income errors can be corrected directly.

Teacher effectiveness: score instability is observable.

Identifying a problem does not establish that latent inference is necessary.

Evidence for interpreting teacher scores

Content: Which aspects of teaching do test scores miss?

Convergent: Do scores agree with other validated evaluations?

Consequential: What happens when schools use these scores?

Specify a finding that would challenge the interpretation or use.

Jacobs & Wallach, 2021, secs. 3.2.2, 3.2.3, 3.2.7

Discussion 3: How would you test the measurement?

Keep the same example and draw on Section 3.

> What evidence would help determine whether the proxy actually measures the intended construct?

Propose one concrete test. What result would cast doubt on the measurement?

Discuss with classmates; record your own response to Question 3.

Jacobs & Wallach, 2021, sec. 3

Shared abstractions shape what gets optimized

Interpretability

A shared vocabulary for discussing the system.

Observable grounding

Connections to variables in the system.

> Competing interests shape construct choices, measurement, and optimization.

Does the evidence support the assurance claim?

Does the decision require this construct estimate?

Do the observations justify this estimate?

Does a measurement mismatch explain the harm?

> Limit the assurance claim to what the evidence supports about the decision.